EV Adoption Trend in India

2014 – 2023 • National Overview

National Trend | State Insights | Infrastructure | Executive Summary

TOTAL EVS REGISTERED

3.6M

Cumulative 2014–2023

CAGR (2014–2023)

107.17%

Compounded annually

LATEST YEAR EVs

1.7M

▲ 65.6% vs 2022

TOTAL PCS (2025)

26,367

▲ 4.6% vs 2024

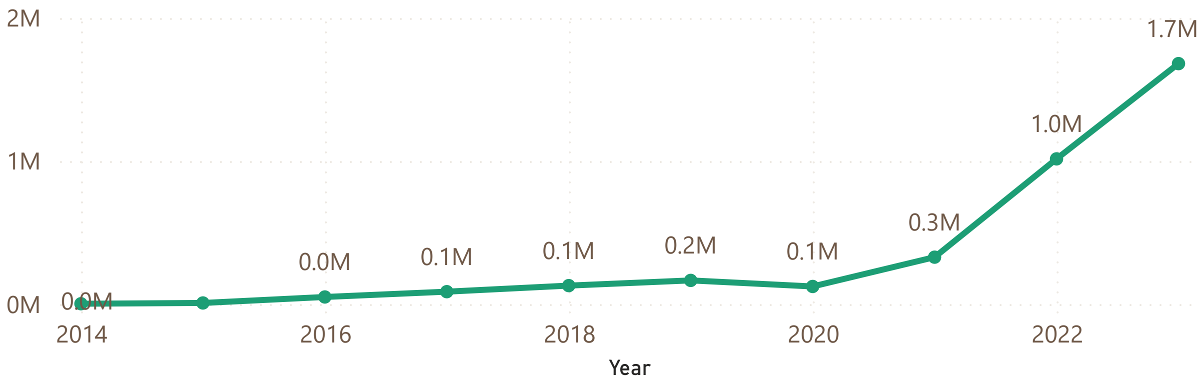
EV vs PCS GROWTH GAP

-65.5pp

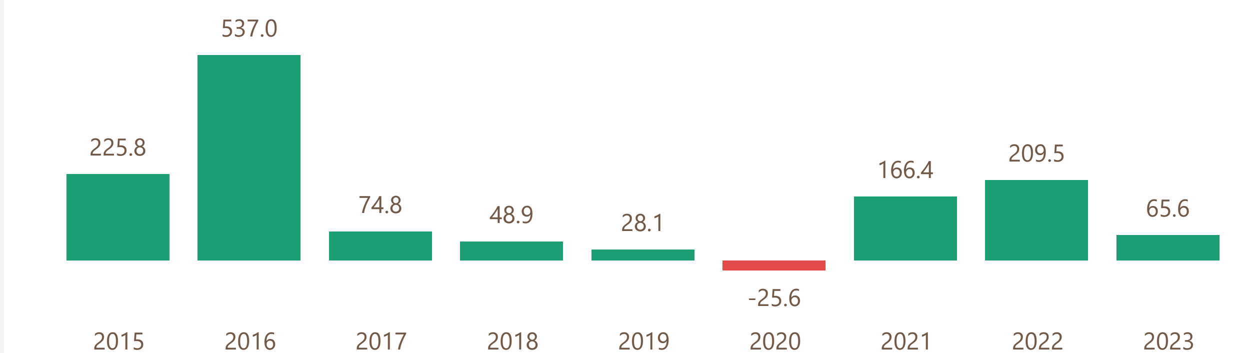
Infra lagging demand

India registered 1.68M EVs in 2023 - a 65.6% jump from 2022, driven by FAME-II subsidies and a 128% surge in public charging stations since 2021.

EV registrations - 2014 to 2023

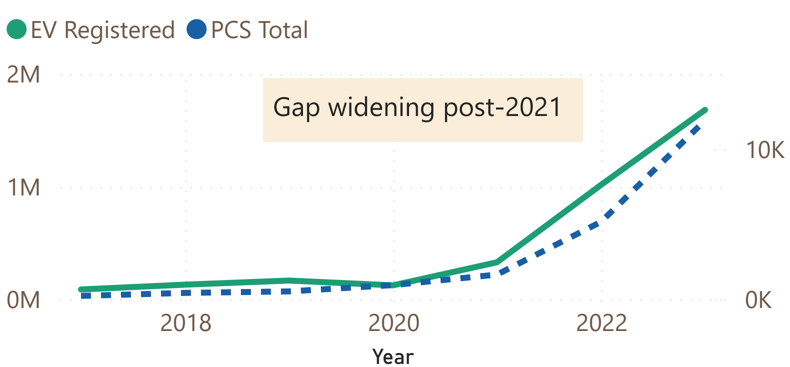


Year-over-Year EV Growth



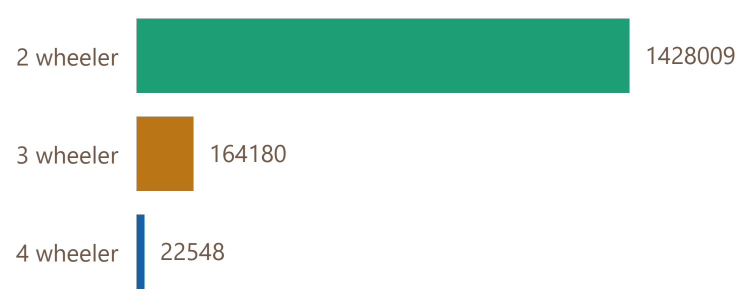
EV growth vs charging infrastructure

Dual-axis — is infra keeping pace with demand?



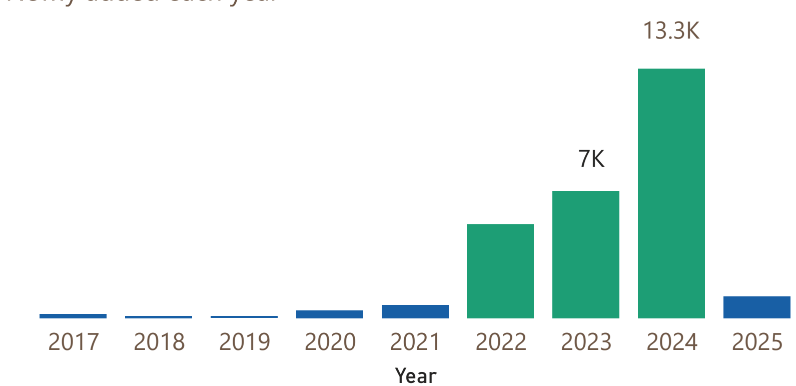
FAME scheme - segment share

EVs supported by vehicle type



PCS growth

Newly added each year



State battleground - EV insights

2020–2023 • State-level performance

TOP EV STATE
Uttar Pradesh
665K total EVs registered

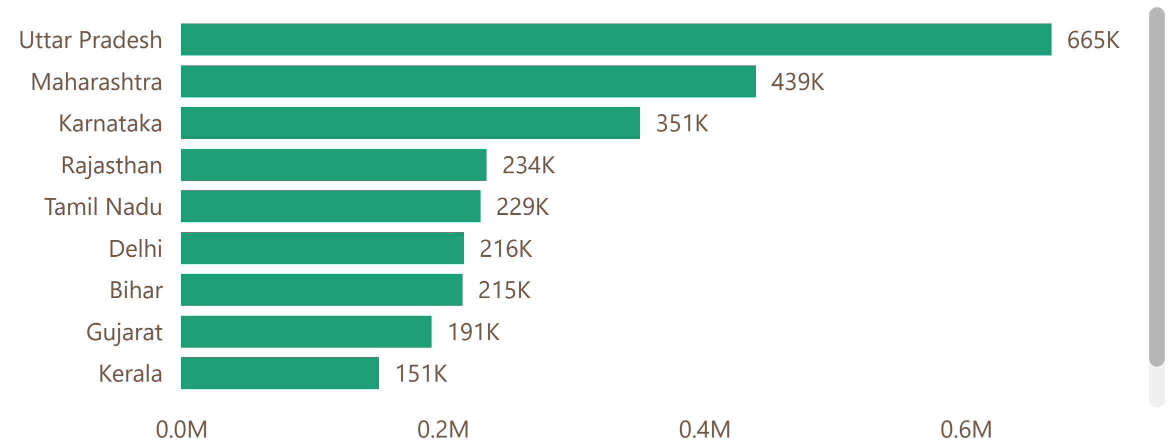
HIGHEST EV SHARE %
Delhi - 7.72%
of all vehicles are EVs

MOST PCS (INFRA LEADER)
Karnataka - 5,765
public charging stations

STATES ABOVE AVG EV SHARE
15 States
Avg national share: 2.66%

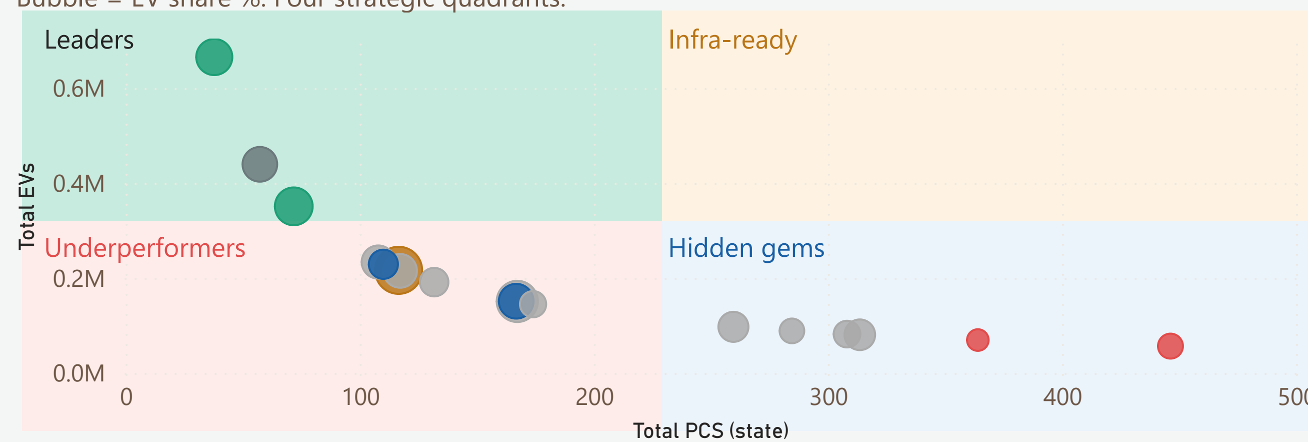
Uttar Pradesh leads in volume (665K EVs) but Delhi leads in penetration (7.72%). Karnataka dominates charging infra with 5,765 stations - yet has the 3rd most EVs, suggesting better readiness.

Top 10 States by EV Registrations - 2023



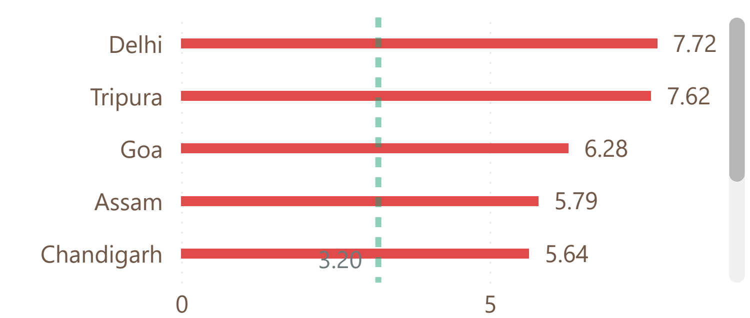
State quadrant analysis — EV adoption vs infra

Bubble = EV share %. Four strategic quadrants.



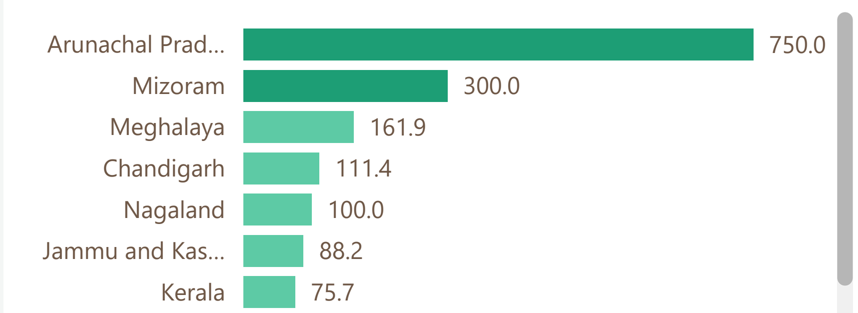
EV share % by state - lollipop chart

States vs national average (3.2%)



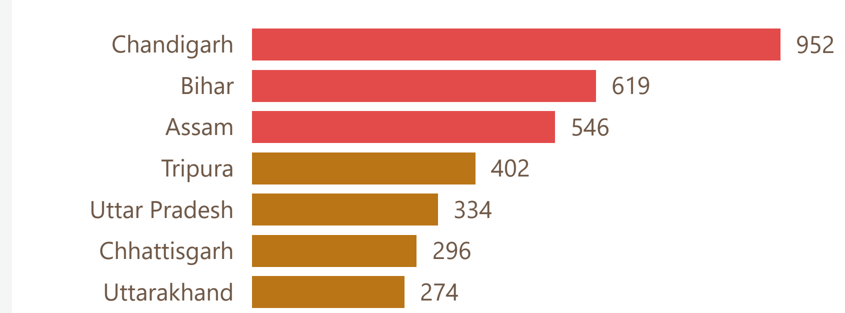
State YoY growth % - 2022 to 2023

Top 8 fastest growing states



PCS vs EV load

EVs per charging station



EV readiness and infrastructure trends

Future outlook • Infrastructure gaps • Segment analysis

HIGH READINESS STATES

15 States

Score above 60 / 100

CRITICAL INFRA OVERLOAD

3 States

Above 500 EVs per station

PCS ACCELERATION RATE

38.37%

Of all PCS added in 2024

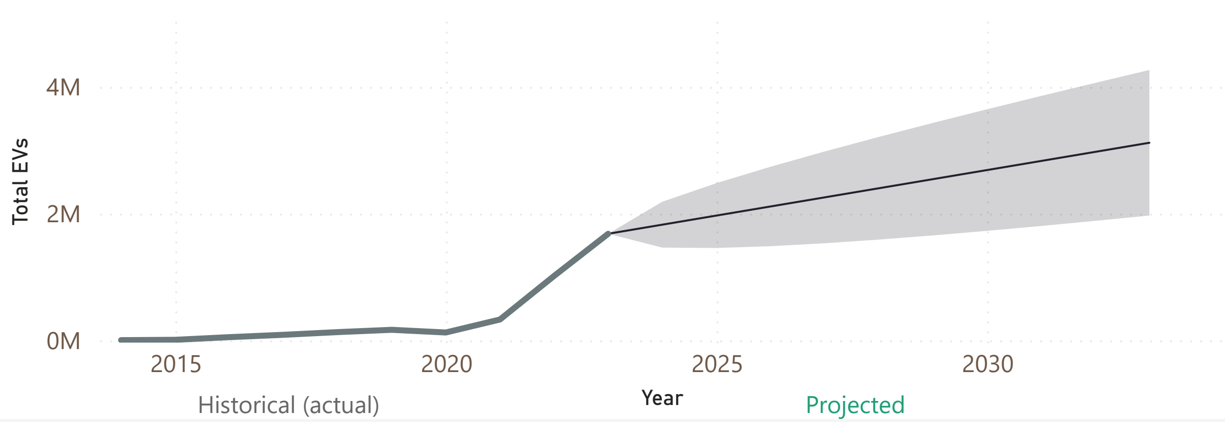
FORECAST EVS BY 2027

2.26M

Range: 1.5M–3.0M

India's EV infra is critically lagging - 3 states have over 500 EVs per charging station. At current PCS growth rates, the infrastructure gap will widen further through 2027 unless targeted investment accelerates in high-load states.

EV adoption forecast - 2024 to 2033

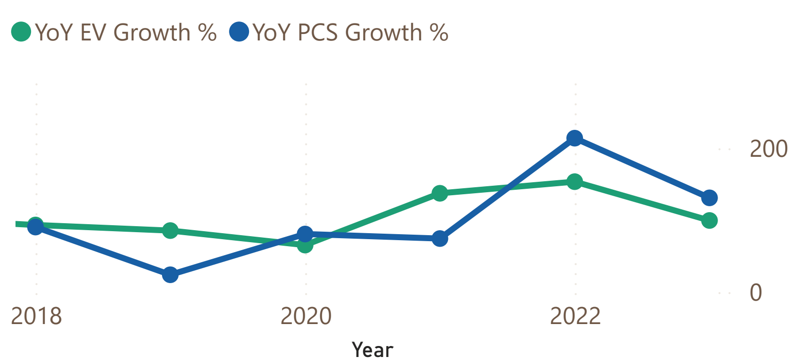


State readiness score — RAG status

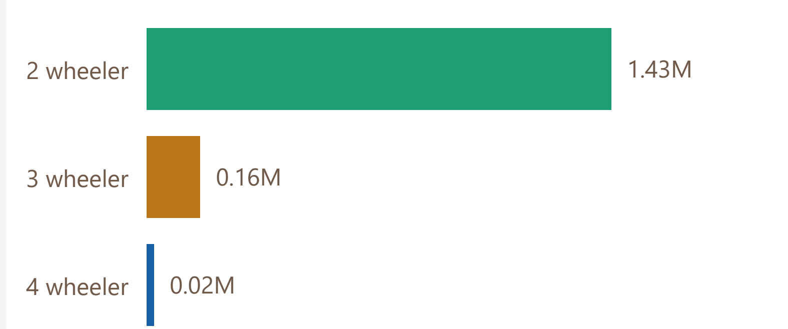
State	EV Readiness Score	Readiness Band
Delhi	90.88	High
Karnataka	79.20	High
Kerala	75.96	High
Maharashtra	75.88	High
Rajasthan	74.40	High
Tamil Nadu	70.80	High
Gujarat	70.12	High
Goa	67.86	High
Uttar Pradesh	67.26	High

YoY EV vs PCS growth - dual trend

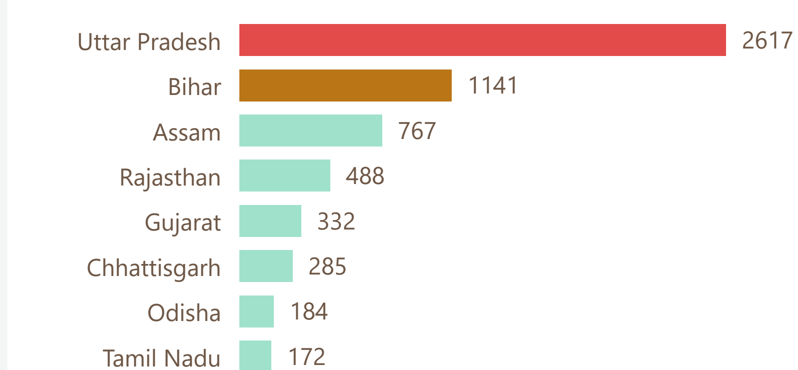
Are charging stations keeping pace?



FAME Scheme — EVs Supported by Segment



PCS gap - states needing most stations



Executive summary — India EV landscape

Consolidated findings • Key insights • Forward outlook

NATIONAL CAGR 2014-2023
107.17%
Fastest growing mobility segment

TOP PERFORMING STATE
Uttar Pradesh
665K EVs - 25% of national total

INFRASTRUCTURE CRISIS INDEX
3 States
Above 500 EVs per charging station

FORECAST 2027
2.26M
Range: 1.5M–3.0M

01 - Growth story

India's EV market grew at 107.17% CAGR from 2014 to 2023. 2022 recorded a 209% surge — the largest single-year jump, driven by FAME-II subsidies and post-COVID demand.

02 - Volume vs penetration

UP leads in volume (665K EVs) but ranks 7th in EV share at 4.3%. Delhi leads penetration at 7.72% - market size and market depth tell very different stories.

03 - Infrastructure crisis

3 states have over 500 EVs per charging station. Chandigarh is most critical at 952 EVs per station. The EV-to-PCS gap has widened sharply since 2021.

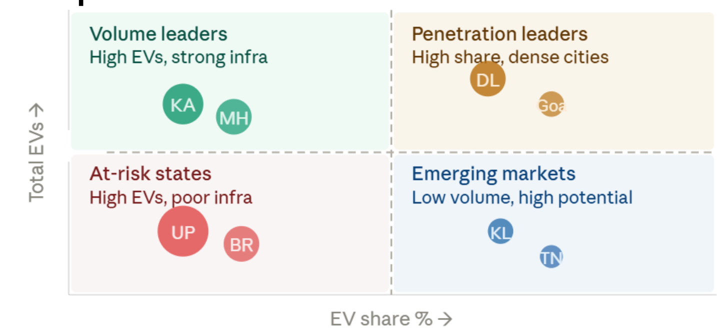
04 - Segment concentration

88.4% of FAME-subsidized EVs are 2-wheelers. 4-wheelers received just 1.4% of support - indicating policy focus on mass mobility over premium adoption.

05 - Readiness gap

15 states score above 60 on the composite readiness index. UP and Bihar hold 40% of national EVs but score below 30 - creating a systemic infrastructure risk.

State performance matrix



Key recommendations

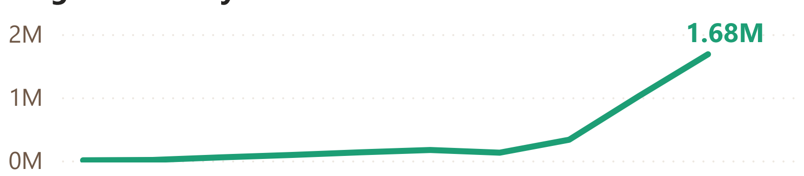
Actions for policymakers and investors

→ Priority PCS investment in UP and Bihar
40% of national EVs are in states with readiness scores below 30

→ Expand FAME support for 4-wheelers
4W segment at only 1.4% of support despite highest unit value

→ Replicate Delhi and Karnataka's model
Both balance high EV share with strong charging infra density

EV growth - 10 year trend



Top 5 states - EV share %

PCS growth — 2017 to 2025